

Exploration of Use Cases within the VEINS platform

Team Pepega Clap

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Project Summary

Description

Students will be tasked with creating various cyber security use cases to simulate attacks against confidentiality, integrity, and availability within the VEINS platform. This platform comprises a vehicle to everything (V2X) networking layer (Omnet++) as well as a simulation layer (Sumo) that allows replication of V2X environments. The work this semester will focus on enumerating possible use cases within a cyber security context and implementing those within VEINS. Please note that some capabilities of VEINS, e.g., message transmission speed, may not allow for certain types of attacks. One of the facets of this project will be identifying potential barriers to implementation and proposing solutions through C++ code analysis.

Motivation

Real world testing is expensive and time consuming. Testing in vehicles is especially challenging to perform due to the specific hardware involved. Such testing can be made easier by simulation as it drastically reduces time and cost. We are interested in using the VEINS simulation platform to explore use cases for testing security in vehicles.

Organization

Deliverables

- Proposals and Literature Review
- Progress Reports and Representation
- Final Report and Presentation
- Documentation and Code delivery

Schedule/Milestones

September 18:	Proposal due
September 24:	Initiate development of use cases proposed
October 2:	Development of use case 1 & 2
October 16:	Midterm (progress) report and presentation
October 30:	Finalization of results, and if time permits, development of a use case 3
November 13:	Developing Final Deliverables and Presentations
TBD:	Final Presentations

Budget

For this project, we expect absolutely no cost. Development will be on the Veins platform, which is completely free to use.

Team Approach and Justification

Use Case 1 - Situational Awareness – Obstacle Collision 2.0

This use case describes a situation where a foreign object is detected on the road ahead in the path of the vehicle. This use case has been broadly covered by “Situational Awareness – Obstacle (SAW – O)”, however, we would like to iterate upon the process. Specifically, we would like to classify obstacles as moving or stationary and implement appropriate courses of action to avoid collision. Examining other use cases such as “Forward Crash Warning (FCW)” or “SA5: Turning Assistant – Vulnerable Road User Avoidance”, we would like to combine these cases into a more generalized case for collision avoidance. We will approach the problem by modeling some of these cases and examining which tactics avoid collisions efficiently. Detection of a foreign object on the road can cause possible on-board navigation systems to reroute the vehicle to other unintended paths. Spoofed objects or obstacles or even hijacked detection systems can lead to a variety of issues including traffic delay, collision, and injury.

Use Case 2 - Loss of Sensor or Network

This use case describes a situation where communication with a sensor on the vehicle is lost. This may be from physical damage to the sensor or a temporary interrupt in the data link. The goal is to explore how a vehicle may act when it is “blind” and what if any default settings are appropriate for such situations. The goal is to create a strategy where the vehicle is still able to operate safely and normally even without sensory feedback. As far as we are aware, there does not exist a use case that covers such a loss and thus we think it is vital for the system to have a failsafe for such a scenario. This use case applies to potential hijacking of sensors and communication systems within vehicles and the possible accidents that may occur in consequence. This use case is directly related to on-board detection systems and the potential of sensor/network failure leading to improper or lack of detection. The failure of these detection and sensory systems could lead to the risk of serious injury, loss of life, and traffic congestion.

Technical Approach

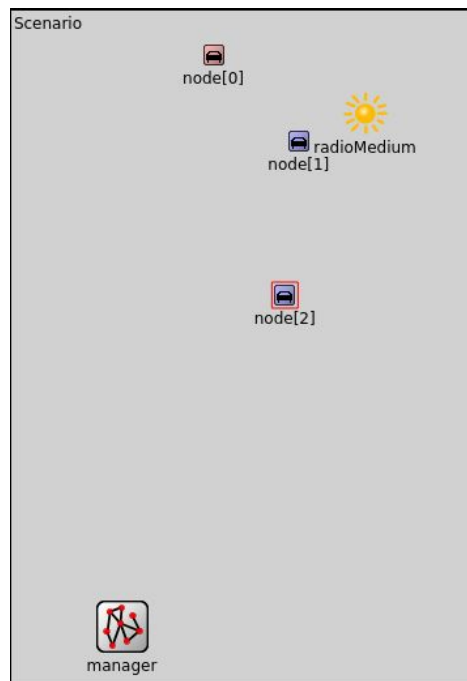
We will be installing the Instant Veins virtual machine which can then be run using any software supporting the Open Virtualization Format. Instant Veins comes preinstalled with three software packages, OMNeT++, SUMO and VEINS, that we will be using to work and interact with the Veins platform.

We will be using C++ to develop use cases for the VEINS platform. We will be modeling use case scenarios by simulating an environment for the vehicle and examining which solutions are most appropriate within certain cases.

Midterm Report

Proposed Use Case 1: Situational Awareness- Obstacle Collision 2.0

To avoid the traffic congestion and incidents that are caused by obstacle objects, we have proposed the situational awareness - obstacle collision 2.0, an extension of use case Situational Awareness - Obstacle. We classified foreign objects as moving or stationary and implemented appropriate courses of action to avoid collision. If an object is detected on the road within the same path of the vehicle, the first car will stop and it will pass a message to all cars that are coming toward within a certain distance. Cars that receive the message will be changed into a cautionary status and be rerouted to an alternative path.



Proposed Use Case 2: Loss of Sensor or Network

Loss of Sensor is a new use case that we are going to approach where communication with a sensor on the vehicle is lost. Our goal is to implement a foundation of security controls to capture when a node loses connection, it is still able to operate safely without sensory feedback. In addition, to avoid any fault or malicious message will be sent to a car, we will be using the voting system to filter out single malicious messages to prevent any fault signal that might cause a car collision or traffic congestion.

Next Plan

We plan to continue trying to understand and be able to figure out all components of the source code so we can make modifications to it. We also need to figure out how to create a custom scenario for the second case. When we have strong understanding in the source code, we will be attempting to apply the method to filter out each fault message to prevent it from being sent to the car.

Literature Review

Secure Data Communication in Autonomous V2X Systems

This paper describes the communication and exchange between the vehicles and the roadside data in the V2X communication systems. The authors are looking forward to giving a solution to protect sensitive information from curious and malicious information cloud administrators. It proposes a solution to the V2X communication system by providing role-based and attribute-based access control to encrypted data and encrypted search over encrypted data to give a higher protection to store sensitive information about the owners and vehicles. The role-based and attribute-based access control mechanisms ensure that each V2X communication network node can access only data items for which the node is authorized and will deny any unauthorized data accesses.

Vehicle-to-Infrastructure (V2I) Message Lexicon

This is a lexicon that was developed to explain the relationships and concepts for V2I messages and identifies the ITS standards where they may be found. It contains the standards for language and definitions for connected vehicles (CV) interface to other vehicles, personal devices and the infrastructure. It explains how V2I messages are constructed and transmitted to convey information between vehicles and the infrastructure. This lexicon was created to assist in the understanding of concepts and activities related to V2I messages and review recent and forthcoming standards that support CV applications.

Intelligent traffic management at intersections supported by V2V and V2I communications

This paper described an intelligent traffic management system that would serve to improve traffic at roundabouts and crossroads. There are two separate algorithms for each scenario. These scenarios were analyzed and shown to have very promising results in improving traffic flow. These were run in a simulator known as the ISR Traffic Simulator. These management algorithms make use of vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communications to provide the algorithms with the necessary information. This management system applies to both autonomous cars and human drivers.

Bibliography

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Biographical Sketches of Investigators

Caleb Hoff

Caleb Hoff is a fourth year student pursuing his masters in Computer Science at the University of Texas at Dallas through the fast-track program. Last year, he visited England's University of Sheffield to study Computer Security and Forensics with Dr. Achim Brucker. While he was there he learned a lot about cyber security fundamentals including SQL injections, XSS attacks, crypto attacks, and static code analysis. He is experienced with a few languages, including: Java, C/C++, Python, C#, and JavaScript. He is especially interested in reverse engineering, binary exploitation, and malware analysis.

Tram Pham

Tram Pham is in her last semester of pursuing her Bachelor's in Computer Science. She is currently a fast track student and working on her master's degree in Computer Science with a specialization in Information Assurance while finishing undergraduate studies. She is also a recipient of the CyberCorps Scholarship for Service (SFS). She has completed related courseworks in computer networks, computer architecture, operating system and database system. She is experienced in Python, Java, C/C++, MIPS and SQL.

Henry Tran

Firstly, Henry Tran is a United States Marine. He is also in his final semester pursuing his Masters of Computer Science, and aims to utilize his degree for the United States Cyber Command. He is also a National Science Foundation Scholarship For Service (SFS) Recipient who aspires to serve his country and work for the US government.

The Marine Corps provided him with communication and leadership abilities as well as analysis and effective writing skills from his MOS. Henry has an interest in penetration testing, binary exploitation, malware analysis, reverse engineering, and network security. He currently participates in the Computer Security Group at UTD, which is an active ground for cyber security. He has experience utilizing nmap, Wireshark, Binary Ninja, nikto, and Metasploit. He is experienced in Python, C/C++, Java, JavaScript, HTML, CSS, and MIPS. Relevant coursework he is taking include Systems Security and Malicious Code Analysis and Software Defined Networks.

James Wei

James Wei is a graduate student studying cyber security at The University of Texas at Dallas. He is a recipient of the CyberCorps Scholarship for Service (SFS) award and the Academic Excellence Scholarship (AES). He participated in a Research Experiences for Undergraduates (REU) program during his undergraduate studies. Specifically, the program focused on software safety and reliability with a heavy base in research and application. During the program, he lead a team to develop a secure online web application with the goal of helping a local charity store their data online.

He chose to study cyber security due to personal interest in the subject and in the first semester of his graduate studies he worked in a coop internship program with L3Harris Technologies. During his internship, he worked in developing a secure system for transferring messages between hardware processors in a radar system.

Dylan Yu

Dylan Yu is a graduate student pursuing a master's degree in cyber security. The coursework he's completed has prepared him with a variety of skills pertaining to network security, reverse engineering, and malware analysis. Through these courses he's learned to use many different technologies including Nmap, Wireshark, Ghidra, IDA Pro, and numerous others. He's well versed in many languages, particularly C, Java, Python, and C#.